Tutorial 2 Solutions

Optimization for Machine Learning (E0 331)

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1. (a) **Norm Composition:** Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined as $f(x) = g(\|x\|)$, where $g : [0, \infty) \to (-\infty, \infty)$ is a convex function.
Show that the proximal operator of f satisfies

$$\operatorname{prox}_f(x) = \begin{cases} \dfrac{\operatorname{prox}_g(\|x\|)}{\|x\|}\, x, & \text{if } x \neq 0, \\ \left\{ u \in \mathbb{R}^n : \|u\| = \operatorname{prox}_g(0) \right\}, & \text{if } x = 0. \end{cases}$$

(b) Using part (a), or otherwise, compute the proximal operator of the Euclidean norm. That is, let

$$f(x) = \lambda \|x\|, \quad \lambda > 0,$$

where $\|\cdot\|$ denotes the Euclidean norm. Derive an explicit expression for $\operatorname{prox}_f(x)$.

Solution.

(a) We compute $\operatorname{prox}_f(x)$ for $x \neq 0$. The proximal problem is

$$\min_{u \in \mathbb{R}^n} \left\{ g(\|u\|) + \frac{1}{2}\|u - x\|^2 \right\}.$$

Expanding,

$$= \min_{u \in \mathbb{R}^n} \left\{ g(\|u\|) + \frac{1}{2}\|u\|^2 - \langle u, x \rangle + \frac{1}{2}\|x\|^2 \right\}.$$

This can be written as

$$= \min_{\alpha \geq 0} \min_{u : \|u\| = \alpha} \left\{ g(\alpha) + \frac{1}{2}\alpha^2 + \frac{1}{2}\|x\|^2 \underbrace{- \langle u, x \rangle}_{\text{only term depending on } u} \right\}.$$

By the Cauchy–Schwarz inequality, the inner minimization is achieved at

$$\boxed{u = \alpha \frac{x}{\|x\|}} \quad (\star)$$

and the corresponding value becomes

$$g(\alpha) + \frac{1}{2}\alpha^2 - \alpha\|x\| + \frac{1}{2}\|x\|^2 = g(\alpha) + \frac{1}{2}(\alpha - \|x\|)^2.$$

Hence,

$$\alpha = \arg\min_{\alpha \geq 0} \left\{ g(\alpha) + \frac{1}{2}(\alpha - \|x\|)^2 \right\} = \mathrm{prox}_g(\|x\|).$$

Therefore, by substituting the value of α in $(\star)$, we obtain

$$\boxed{\mathrm{prox}_f(x) = \frac{\mathrm{prox}_g(\|x\|)}{\|x\|}\, x}$$

Now consider $x = 0$. By definition,

$$\mathrm{prox}_f(0) = \arg\min_{u \in \mathbb{R}^n} \left\{ g(\|u\|) + \frac{1}{2}\|u\|^2 \right\}.$$

Let $w = \|u\|$. Then the problem reduces to

$$\min_{w \geq 0} \left\{ g(w) + \frac{1}{2} w^2 \right\}.$$

Hence $w^* = \mathrm{prox}_g(0)$, and the set of minimizers is

$$\boxed{\mathrm{prox}_f(0) = \left\{ u \in \mathbb{R}^n : \|u\| = \mathrm{prox}_g(0) \right\}}$$

(b) Let $f(x) = \lambda\|x\|$, $\lambda > 0$. Then $f(x) = g(\|x\|)$ with

$$g(t) = \begin{cases} \lambda t, & t \geq 0, \\ +\infty, & t < 0. \end{cases}$$

The proximal operator of g is

$$\mathrm{prox}_g(t) = [t - \lambda]_+ = \max(t - \lambda, 0).$$

In particular, $\mathrm{prox}_g(0) = 0$.
Using part (a), for $x \neq 0$,

$$\mathrm{prox}_f(x) = \frac{[\|x\| - \lambda]_+}{\|x\|}\, x,$$

and for $x = 0$,

$$\mathrm{prox}_f(0) = 0.$$

Thus,

$$\mathrm{prox}_f(x) = \begin{cases} \left(1 - \dfrac{\lambda}{\|x\|}\right) x, & \|x\| > \lambda, \\ 0, & \|x\| \leq \lambda. \end{cases}$$

Equivalently, this can be written compactly as

$$\boxed{\mathrm{prox}_{\lambda\|\cdot\|}(x) = \left(1 - \frac{\lambda}{\max\{\|x\|, \lambda\}}\right) x}$$

2. **Huber Loss:** The Huber loss is a widely used loss function in machine learning due to its robustness to outliers while maintaining smoothness. It is commonly used in regression problems, deep learning, reinforcement learning (e.g., Q-learning), and computer vision tasks.

   In this question, we show that the Huber loss arises as the Moreau envelope of the Euclidean norm and is $\frac{1}{\mu}$-smooth.

   (a) Let $f(x) = \|x\|$ be the Euclidean norm on $\mathbb{R}^n$. Show that its Moreau envelope with parameter $\mu > 0$ is given by the Huber function

$$H_\mu(x) = \begin{cases} \dfrac{1}{2\mu}\|x\|^2, & \text{if } \|x\| \le \mu, \\ \|x\| - \dfrac{\mu}{2}, & \text{if } \|x\| > \mu. \end{cases}$$

   (b) Show that H_μ is $\frac{1}{\mu}$-smooth.

Solution.

We recall that for a convex function $f : \mathbb{R}^n \to \mathbb{R}$ and $\mu > 0$, the Moreau envelope is defined as

$$M_\mu f(x) = \min_{u \in \mathbb{R}^n} \left\{ f(u) + \frac{1}{2\mu}\|x - u\|^2 \right\}.$$

It is known that:

- $M_\mu f$ is convex and real-valued,
- $M_\mu f$ is $\frac{1}{\mu}$-smooth.

(a) Let $f(x) = \|x\|$. From the previous problem, we know that

$$\operatorname{prox}_{\mu f}(x) = \left(1 - \frac{\mu}{\max\{\|x\|, \mu\}}\right) x.$$

Using the identity for the Moreau envelope,

$$M_\mu f(x) = f\big(\operatorname{prox}_{\mu f}(x)\big) + \frac{1}{2\mu}\|x - \operatorname{prox}_{\mu f}(x)\|^2,$$

we compute it case-wise.

Case 1: $\|x\| \le \mu$**.**

Then $\operatorname{prox}_{\mu f}(x) = 0$, so

$$M_\mu f(x) = 0 + \frac{1}{2\mu}\|x\|^2 = \frac{1}{2\mu}\|x\|^2.$$

Case 2: $\|x\| > \mu$**.**

Then

$$\operatorname{prox}_{\mu f}(x) = \left(1 - \frac{\mu}{\|x\|}\right) x.$$

Hence

$$\left\|\operatorname{prox}_{\mu f}(x)\right\| = \|x\| - \mu, \quad \|x - \operatorname{prox}_{\mu f}(x)\| = \mu.$$

Therefore,

$$M_\mu f(x) = (\|x\| - \mu) + \frac{1}{2\mu}\mu^2 = \|x\| - \frac{\mu}{2}.$$

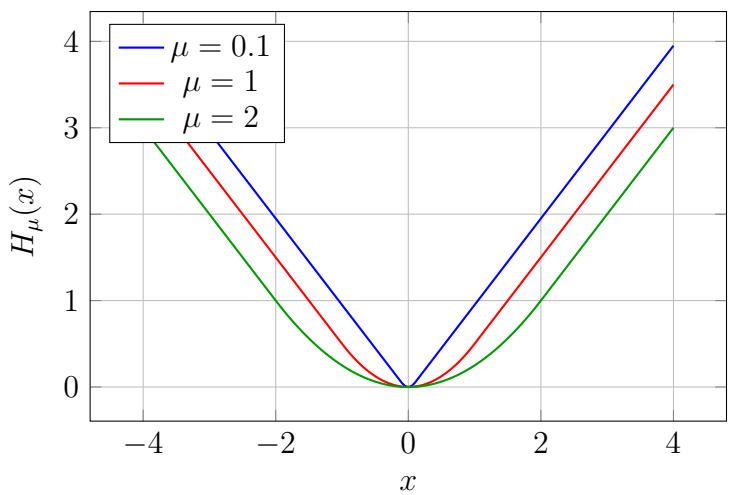
Thus,

$$\boxed{M_\mu \|\cdot\|(x) = H_\mu(x)}$$

where

$$H_\mu(x) = \begin{cases} \frac{1}{2\mu}\|x\|^2, & \|x\| \le \mu, \\ \|x\| - \frac{\mu}{2}, & \|x\| > \mu. \end{cases}$$

(b) Since the Moreau envelope $M_\mu f$ of any convex function is $\frac{1}{\mu}$-smooth, it follows directly that H_μ is $\frac{1}{\mu}$-smooth.



Note: From the above plot, we observe how the Huber loss behaves for different values of μ. The parameter μ controls the transition between quadratic and linear behavior. For small μ, the function behaves more like the absolute value (linear), while for larger μ, it exhibits a wider quadratic region near the origin. Thus, μ determines whether the loss is more sensitive (quadratic) or more robust (linear) to deviations.

3. **Prox of Moreau Envelope:** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function, and let $\mu > 0$. Show that for any $x \in \mathbb{R}^n$,

$$\operatorname{prox}_{M_\mu f}(x) = x + \frac{1}{\mu+1}\left(\operatorname{prox}_{(\mu+1)f}(x) - x\right).$$

Using this result, compute the proximal operator of the Huber function H_μ, defined as the Moreau envelope of the Euclidean norm.

Solution.

First, observe that

$$\min_u \; M_\mu f(u) + \frac{1}{2}\|u-x\|^2 = \min_u \; \min_y \left\{ f(y) + \frac{1}{2\mu}\|u-y\|^2 + \frac{1}{2}\|u-x\|^2 \right\}.$$

Exchanging the order of minimization,

$$= \min_y \; \min_u \left\{ f(y) + \frac{1}{2\mu}\|u-y\|^2 + \frac{1}{2}\|u-x\|^2 \right\}.$$

For fixed y, the inner minimization in u is quadratic. Setting the gradient to zero,

$$\frac{1}{\mu}(u-y) + (u-x) = 0,$$

which gives

$$u = u_\mu = \frac{\mu x + y}{\mu+1}.$$

Substituting back,

$$\begin{aligned} &f(y) + \frac{1}{2\mu}\|u_\mu - y\|^2 + \frac{1}{2}\|u_\mu - x\|^2 \\ &= f(y) + \frac{1}{2(\mu+1)}\|x-y\|^2. \end{aligned}$$

Thus, the problem reduces to

$$\min_y \left\{ f(y) + \frac{1}{2(\mu+1)}\|x-y\|^2 \right\},$$

whose solution is

$$y = \operatorname{prox}_{(\mu+1)f}(x).$$

Therefore,

$$\boxed{\operatorname{prox}_{M_\mu f}(x) = \frac{1}{\mu+1}\left(\mu x + \operatorname{prox}_{(\mu+1)f}(x)\right)}$$

Now, consider $f(x) = \lambda H_\mu(x)$, where $H_\mu = M_\mu g$ with $g(x) = \|x\|$. Then

$$\operatorname{prox}_{\lambda H_\mu}(x) = \operatorname{prox}_{\lambda M_\mu g}(x).$$

Using the previous result,

$$\operatorname{prox}_{\lambda H_\mu}(x) = x + \frac{\lambda}{\mu + \lambda}\left(\operatorname{prox}_{(\mu+\lambda)g}(x) - x\right).$$

From the proximal operator of the Euclidean norm,

$$\operatorname{prox}_{(\mu+\lambda)g}(x) = \left(1 - \frac{\mu + \lambda}{\max\{\|x\|, \mu + \lambda\}}\right) x.$$

Substituting,

$$\operatorname{prox}_{\lambda H_\mu}(x) = x + \frac{\lambda}{\mu + \lambda}\left[\left(1 - \frac{\mu + \lambda}{\max\{\|x\|, \mu + \lambda\}}\right) x - x\right].$$

After simplification,

$$\boxed{\operatorname{prox}_{\lambda H_\mu}(x) = \left(1 - \frac{\lambda}{\max\{\|x\|, \mu + \lambda\}}\right) x}$$

4. **Mirror Descent:** We now consider an important algorithm called *Mirror Descent*, which generalizes projected subgradient methods by replacing Euclidean geometry with a more general geometry.

   To understand the role of the Euclidean norm, consider the reformulation

$$x_{k+1} = \arg\min_{x \in C} \left\{ f(x_k) + \langle f'(x_k), x - x_k \rangle + \frac{1}{2t_k} \|x - x_k\|^2 \right\}.$$

   This shows that x_{k+1} is obtained by minimizing a linear approximation of f together with a quadratic proximity term. The key idea is to replace the Euclidean distance in this formulation with a more general divergence, thereby inducing a non-Euclidean geometry tailored to the problem.

   We now introduce the central object in mirror descent.

   Bregman Divergence. Let $\omega : \mathbb{R}^n \to \mathbb{R}$ be differentiable and σ-strongly convex. Define

$$B_\omega(x, y) = \omega(x) - \omega(y) - \langle \nabla\omega(y), x - y \rangle.$$

   Basic properties:

   (a) $B_\omega(x, y) \geq \frac{\sigma}{2} \|x - y\|^2$ for all $x, y \in \mathbb{R}^n$.

   (b) $B_\omega(x, y) \geq 0$, and $B_\omega(x, y) = 0$ if and only if $x = y$.

   We now present the Mirror Descent algorithm.

   Mirror Descent Method

   Initialization: pick $x_0 \in C$.

   For $k = 0, 1, 2, \ldots$:

   - Choose a stepsize $t_k > 0$ and $f'(x_k) \in \partial f(x_k)$.
   - Update

$$x_{k+1} = \arg\min_{x \in C} \left\{ \langle t_k f'(x_k) - \nabla\omega(x_k), x \rangle + \omega(x) \right\}.$$

   To understand the algorithm better we now consider a concrete example.

   Example (Negative Entropy over the Unit Simplex).

   Let $C = \Delta^n$ and

$$\omega(x) = \begin{cases} \sum_{i=1}^n x_i \log x_i, & x \in \mathbb{R}^n_+, \\ \infty, & \text{otherwise.} \end{cases}$$

Show that the update satisfies

$$\boxed{(x_{k+1})_i = \frac{(x_k)_i \, e^{-t_k f_i'(x_k)}}{\sum_{j=1}^n (x_k)_j \, e^{-t_k f_j'(x_k)}}, \quad i = 1, 2, \ldots, n.}$$

We now turn to the convergence analysis. To this end, we first establish three fundamental lemmas that will serve as the main technical tools. We present them sequentially.

Lemma 1 (Three-point identity).

Let $\omega : \mathbb{R}^n \to \mathbb{R}$ be a differentiable convex function. Then, for all $a, b, c \in \mathbb{R}^n$,

$$\langle \nabla\omega(b) - \nabla\omega(a),\, c - a \rangle = B_\omega(c, a) + B_\omega(a, b) - B_\omega(c, b).$$

Lemma 2 (Non-Euclidean prox inequality).

Let $\omega : \mathbb{R}^n \to \mathbb{R}$ be a differentiable convex function that is σ-strongly convex ($\sigma > 0$). Let $\psi : \mathbb{R}^n \to (-\infty, \infty]$ be a convex function.

Let $b \in \mathbb{R}^n$, and define

$$a = \arg\min_{x \in \mathbb{R}^n} \{\psi(x) + B_\omega(x, b)\}.$$

Then $a \in \mathbb{R}^n$, and for all $u \in \mathbb{R}^n$,

$$\langle \nabla\omega(b) - \nabla\omega(a), u - a \rangle \le \psi(u) - \psi(a).$$

Lemma 3 (Fundamental inequality).

Assume that f is convex, $C \subseteq \mathbb{R}^n$ is a closed convex set, and $\omega : \mathbb{R}^n \to \mathbb{R}$ is differentiable and σ-strongly convex ($\sigma > 0$). Let $\{x_k\}_{k \ge 0}$ be the sequence generated by the mirror descent method with positive stepsizes $\{t_k\}_{k \ge 0}$. Then, for any minimizer x^* and for all $k \ge 0$,

$$t_k\big(f(x_k) - f_{\text{opt}}\big) \le B_\omega(x^*, x_k) - B_\omega(x^*, x_{k+1}) + \frac{t_k^2}{2\sigma} \|f'(x_k)\|_*^2.$$

Main Theorem.

Assume $\|f'(x)\|_* \le L_f$ for all $x \in C$, and $\Theta(x_0) \ge \max_{x \in C} B_\omega(x, x_0)$.

Then for any $N \ge 0$,

$$\boxed{f_N^{\text{best}} - f_{\text{opt}} \le \frac{\Theta(x_0) + \dfrac{L_f^2}{2\sigma} \sum_{k=0}^N t_k^2}{\sum_{k=0}^N t_k}}$$

where

$$f_N^{\text{best}} = \min_{0 \le k \le N} f(x_k).$$

Here, $\Theta(x_0)$ denotes a constant satisfying $\Theta(x_0) \geq \max_{x \in C} B_\omega(x, x_0)$, which measures the size of the feasible set C under the geometry induced by ω.

We now address the choice of stepsizes. The question is: *how should the stepsizes* $\{t_k\}$ *be chosen to minimize the bound?*

To do so we fix the number of iterations N and determine the stepsizes that minimize the right-hand side of the bound. This leads to the optimization problem

Lemma 4 (Optimal Stepsizes).

Let $\alpha, \beta > 0$, and consider the optimization problem

$$\min_{t_1,\ldots,t_m>0} \frac{\alpha + \beta \sum_{k=1}^m t_k^2}{\sum_{k=1}^m t_k}.$$

Then the optimal solution is given by

$$t_k = \sqrt{\frac{\alpha}{\beta m}}, \quad k = 1, 2, \ldots, m,$$

and the optimal value is

$$\boxed{\frac{2\sqrt{\alpha\beta}}{\sqrt{m}}.}$$

Using Lemma 4 with $\alpha = \Theta(x_0)$, $\beta = \dfrac{L_f^2}{2\sigma}$, and $m = N + 1$, we conclude that the minimum of the right-hand side of the boxed inequality in the main theorem is attained at

$$t_k = \sqrt{\frac{2\sigma\Theta(x_0)}{L_f^2(N+1)}}, \quad k = 0, 1, \ldots, N,$$

and the resulting bound becomes

$$\boxed{f_N^{\text{best}} - f_{\text{opt}} \leq \frac{\sqrt{2\Theta(x_0)L_f^2/\sigma}}{\sqrt{N+1}}.}$$

In particular, we obtain the convergence rate

$$f_N^{\text{best}} - f_{\text{opt}} = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right).$$

Solution.

Basic properties.

Part (a) follows by the first-order characterization of strongly convex functions. Part (b) is a direct consequence of part (a).

Example: Negative Entropy over the Simplex.

Let

$$\Delta^n = \left\{ x \in \mathbb{R}^n_+ : \sum_{i=1}^n x_i = 1 \right\}, \quad \Delta^n_+ = \left\{ x \in \mathbb{R}^n_{++} : \sum_{i=1}^n x_i = 1 \right\}.$$

The associated Bregman distance is given for any $x \in \Delta^n$ and $y \in \Delta^n_+$ by

$$\begin{aligned} B_\omega(x, y) &= \sum_{i=1}^n x_i \log x_i - \sum_{i=1}^n y_i \log y_i - \sum_{i=1}^n (\log y_i + 1)(x_i - y_i) \\ &= \sum_{i=1}^n x_i \log\left(\frac{x_i}{y_i}\right), \end{aligned}$$

which is the so-called Kullback–Leibler divergence.

The mirror descent update becomes (where $f'(x_k) \in \partial f(x_k)$ and $f'_i(x_k)$ denotes the ith component of $f'(x_k)$)

$$\boxed{x_{k+1} = \arg\min_{x \in \Delta^n} \left\{ \sum_{i=1}^n \left(t_k f'_i(x_k) - 1 - \log((x_k)_i)\right) x_i + \sum_{i=1}^n x_i \log x_i \right\}}$$

In order to solve the above problem, we need a technique called KKT conditions. Let us take a small detour and review them.

$$\left.\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = a_i^\top x - b_i = 0, \quad \forall i \in [m], \\ & g_j(x) \le 0, \quad \forall j \in [k] \end{aligned}\right\} \quad \text{(CP)}$$

For the convex program (CP), a point x^* is optimal if and only if there exist multipliers $\{\lambda_i\}_{i=1}^m$ and $\{\mu_j\}_{j=1}^k$ such that the following KKT conditions are satisfied:

$$\boxed{\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla h_i(x^*) + \sum_{j=1}^k \mu_j \nabla g_j(x^*) = 0} \quad \text{(stationarity)}$$

$$\boxed{h_i(x^*) = 0 \ \ \forall i \in [m], \quad g_j(x^*) \le 0 \ \ \forall j \in [k]} \quad \text{(feasibility)}$$

$$\boxed{\mu_j \ge 0 \ \ \forall j \in [k], \quad \mu_j g_j(x^*) = 0 \ \ \forall j \in [k]} \quad \text{(complementary slackness)}$$

Now let us quickly apply the KKT conditions to solve the problem.

The optimization problem is

$$\min_{x\in\Delta^n} \sum_{i=1}^{n} \left(t_k f_i'(x_k) - 1 - \log((x_k)_i)\right) x_i + \sum_{i=1}^{n} x_i \log x_i.$$

The Lagrangian is

$$\mathcal{L}(x,\lambda,\mu) = \sum_{i=1}^{n} x_i \log x_i + \sum_{i=1}^{n} \left(t_k f_i'(x_k) - 1 - \log((x_k)_i)\right) x_i + \lambda \left(\sum_{i=1}^{n} x_i - 1\right) - \sum_{i=1}^{n} \mu_i x_i.$$

Now compute $\frac{\partial \mathcal{L}}{\partial x_i} = 0$:

$$\log x_i + 1 + t_k f_i'(x_k) - 1 - \log((x_k)_i) + \lambda - \mu_i = 0.$$

Thus,

$$\log x_i = \log((x_k)_i) - t_k f_i'(x_k) - \lambda + \mu_i.$$

From complementary slackness, $\mu_i x_i = 0$. Since $x_i > 0$ for all i, it follows that

$$\mu_i = 0 \quad \forall i.$$

Hence,

$$\log x_i = \log((x_k)_i) - t_k f_i'(x_k) - \lambda.$$

Exponentiating,

$$x_i = (x_k)_i \, e^{-t_k f_i'(x_k)} e^{-\lambda}.$$

Using the feasibility condition $\sum_{i=1}^{n} x_i = 1$, we obtain

$$e^{-\lambda} = \frac{1}{\sum_{j=1}^{n} (x_k)_j e^{-t_k f_j'(x_k)}}.$$

Substituting back,

$$\boxed{(x_{k+1})_i = \frac{(x_k)_i \, e^{-t_k f_i'(x_k)}}{\sum_{j=1}^{n} (x_k)_j \, e^{-t_k f_j'(x_k)}}}$$

This update is known as the *exponential gradient descent* method.

<u>Proof of Lemma 1.</u>

By definition of the Bregman divergence,

$$\begin{aligned} B_\omega(c,a) &= \omega(c) - \omega(a) - \langle \nabla\omega(a), c-a\rangle, \\ B_\omega(a,b) &= \omega(a) - \omega(b) - \langle \nabla\omega(b), a-b\rangle, \\ B_\omega(c,b) &= \omega(c) - \omega(b) - \langle \nabla\omega(b), c-b\rangle. \end{aligned}$$

Summing the first two expressions and subtracting the third, we obtain

$$\begin{aligned}&B_\omega(c,a)+B_\omega(a,b)-B_\omega(c,b)\\&=-\langle\nabla\omega(a),c-a\rangle-\langle\nabla\omega(b),a-b\rangle+\langle\nabla\omega(b),c-b\rangle.\end{aligned}$$

Rearranging terms,

$$\langle\nabla\omega(b),c-b\rangle-\langle\nabla\omega(b),a-b\rangle=\langle\nabla\omega(b),c-a\rangle.$$

Thus,

$$B_\omega(c,a)+B_\omega(a,b)-B_\omega(c,b)=\langle\nabla\omega(b)-\nabla\omega(a),c-a\rangle,$$

which proves the result.

<u>Proof of Lemma 2.</u>

Using the definition of B_ω, we can rewrite the problem as

$$a=\arg\min_x \underbrace{\{\psi(x)-\langle\nabla\omega(b),x\rangle+\omega(x)\}}_{:=\phi(x)}.$$

By Fermat's rule,

$$0\in\partial\phi(a).$$

Thus, there exists $\psi'(a)\in\partial\psi(a)$ such that

$$\psi'(a)+\nabla\omega(a)-\nabla\omega(b)=0.$$

Hence,

$$\langle\nabla\omega(b)-\nabla\omega(a),u-a\rangle=\langle\psi'(a),u-a\rangle\le\psi(u)-\psi(a),$$

where the last inequality follows from the subgradient inequality, i.e.,

$$\psi(u)\ge\psi(a)+\langle\psi'(a),u-a\rangle.$$

<u>Proof of Lemma 3.</u>

By the update formula and Lemma 2 (non-Euclidean prox inequality), with $b=x_k$ and

$$\psi(x)=t_k\langle f'(x_k),x\rangle+\delta_C(x),$$

we obtain for any $u\in C$,

$$\langle\nabla\omega(x_k)-\nabla\omega(x_{k+1}),u-x_{k+1}\rangle\le t_k\langle f'(x_k),u-x_{k+1}\rangle.$$

Using the three-point identity (Lemma 1) with $a=x_{k+1}$, $b=x_k$, and $c=u$, we have

$$\langle\nabla\omega(x_k)-\nabla\omega(x_{k+1}),u-x_{k+1}\rangle=B_\omega(u,x_{k+1})+B_\omega(x_{k+1},x_k)-B_\omega(u,x_k).$$

Combining the above two relations gives

$$B_\omega(u, x_{k+1}) + B_\omega(x_{k+1}, x_k) - B_\omega(u, x_k) \leq t_k \langle f'(x_k), u - x_{k+1} \rangle.$$

Rearranging,

$$t_k \langle f'(x_k), x_k - u \rangle \leq B_\omega(u, x_k) - B_\omega(u, x_{k+1}) - B_\omega(x_{k+1}, x_k) + t_k \langle f'(x_k), x_k - x_{k+1} \rangle.$$

Since ω is σ-strongly convex, we have

$$B_\omega(x_{k+1}, x_k) \geq \frac{\sigma}{2} \|x_{k+1} - x_k\|^2.$$

Thus,

$$t_k \langle f'(x_k), x_k - u \rangle \leq B_\omega(u, x_k) - B_\omega(u, x_{k+1}) - \frac{\sigma}{2} \|x_{k+1} - x_k\|^2 + t_k \langle f'(x_k), x_k - x_{k+1} \rangle.$$

Now apply Fenchel's inequality:

$$\langle a, b \rangle \leq \frac{1}{2\sigma} \|a\|_*^2 + \frac{\sigma}{2} \|b\|^2,$$

with $a = t_k f'(x_k)$ and $b = x_k - x_{k+1}$, to obtain

$$t_k \langle f'(x_k), x_k - x_{k+1} \rangle \leq \frac{t_k^2}{2\sigma} \|f'(x_k)\|_*^2 + \frac{\sigma}{2} \|x_{k+1} - x_k\|^2.$$

Substituting back,

$$t_k \langle f'(x_k), x_k - u \rangle \leq B_\omega(u, x_k) - B_\omega(u, x_{k+1}) + \frac{t_k^2}{2\sigma} \|f'(x_k)\|_*^2.$$

Finally, setting $u = x^*$ and using the subgradient inequality,

$$f(x_k) - f_{\text{opt}} \leq \langle f'(x_k), x_k - x^* \rangle,$$

we obtain

$$\boxed{t_k (f(x_k) - f_{\text{opt}}) \leq B_\omega(x^*, x_k) - B_\omega(x^*, x_{k+1}) + \frac{t_k^2}{2\sigma} \|f'(x_k)\|_*^2} \quad (*)$$

<u>Main theorem.</u>

Summing equation $(*)$ for $k = 0$ to N, and using the assumption $\|f'(x)\|_* \leq L_f$ for all $x \in C$, and $\Theta(x_0) \geq \max_{x \in C} B_\omega(x, x_0)$, we get

$$\sum_{k=0}^{N} t_k (f(x_k) - f_{\text{opt}}) \leq B_\omega(x^*, x_0) + \frac{L_f^2}{2\sigma} \sum_{k=0}^{N} t_k^2.$$

Now, using the definition of $f_N^{\text{best}} = \min_{0\le k\le N} f(x_k)$, we obtain

$$\left(\sum_{k=0}^{N} t_k\right)\left(f_N^{\text{best}} - f_{\text{opt}}\right) \le \sum_{k=0}^{N} t_k(f(x_k) - f_{\text{opt}}).$$

Combining the above inequalities,

$$\boxed{f_N^{\text{best}} - f_{\text{opt}} \le \frac{\Theta(x_0) + \dfrac{L_f^2}{2\sigma}\sum_{k=0}^{N} t_k^2}{\sum_{k=0}^{N} t_k}}.$$

<u>Lemma 4 (Optimal stepsizes).</u>

Denote the objective function by

$$J(t) = \frac{\alpha + \beta\sum_{k=1}^{m} t_k^2}{\sum_{k=1}^{m} t_k}, \quad \text{where } t = (t_1, \dots, t_m)^\top.$$

Note that J is a permutation symmetric function, meaning that $J(t) = J(Pt)$ for any permutation matrix $P \in \Lambda_m$.

A consequence of this observation is that if the problem admits an optimal solution, then it necessarily has an optimal solution in which all components are equal. To see this, let t^* be an optimal solution and let $P \in \Lambda_m$. Since $J(Pt^*) = J(t^*)$, it follows that Pt^* is also optimal.

Since J is convex over the positive orthant, the average over all permutations is also optimal. In particular,

$$\frac{1}{m!}\sum_{P\in\Lambda_m} Pt^* = \frac{1}{m}\begin{pmatrix} e^\top t^* \\ \vdots \\ e^\top t^* \end{pmatrix}$$

is also an optimal solution. where $e = (1, \dots, 1)^\top$.

Hence, there exists an optimal solution in which all components are equal, i.e.,

$$t_1 = t_2 = \cdots = t_m = t.$$

Therefore, the problem reduces to

$$\min_{t>0} \frac{\alpha + \beta m t^2}{mt}.$$

The minimizer is

$$t = \sqrt{\frac{\alpha}{\beta m}},$$

and the optimal value is

$$\frac{2\sqrt{\alpha\beta}}{\sqrt{m}}.$$

Substituting $\alpha = \Theta(x_0)$, $\beta = \frac{L_f^2}{2\sigma}$, $m = N+1$, we obtain

$$t_k = \sqrt{\frac{2\sigma\Theta(x_0)}{L_f^2(N+1)}}.$$

Finally,

$$\boxed{f_N^{\text{best}} - f_{\text{opt}} \leq \frac{\sqrt{2\Theta(x_0)L_f^2/\sigma}}{\sqrt{N+1}}}$$

5. **Heavy Ball Method:** We have seen several first-order methods so far. A natural question arises:

Can we obtain faster convergence rates (e.g., faster linear rates for strongly convex functions or faster sublinear rates for general convex functions) while still using only first-order information?

The answer is **yes**. The key idea is to introduce *momentum*, where the search direction depends not only on the current gradient $-\nabla f(x_k)$ but also on the previous step, thereby incorporating information from earlier iterations.

Several important methods use this idea, including:

- Heavy Ball method,
- Nesterov's accelerated gradient method.

In this tutorial, we focus on the **Heavy Ball method**, originally proposed by Polyak (1964).

Algorithm. The Heavy Ball method is given by

$$\boxed{x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta(x_k - x_{k-1}).}$$

We analyze the convergence of this method for the special case of strongly convex quadratic functions.

Consider

$$f(x) = \frac{1}{2} x^\top Q x,$$

where Q is symmetric positive definite with eigenvalues

$$0 < \ell = \lambda_n \leq \lambda_{n-1} \leq \cdots \leq \lambda_2 \leq \lambda_1 = L.$$

(a) Show that the Heavy Ball iteration can be written in terms of the eigenvalues of Q and analyze the resulting scalar recursion along each eigendirection.

(b) Show that appropriate choices of α and β (depending on ℓ and L) yield a linear convergence rate.

(c) Derive that the optimal rate satisfies

$$\rho \approx 1 - \frac{2}{\sqrt{\kappa}},$$

where $\kappa = \frac{L}{\ell}$ is the condition number.

Solution.

(a) We consider the Heavy Ball iteration

$$x_{k+1} = x_k - \alpha Q x_k + \beta(x_k - x_{k-1}),$$

where α, β are fixed constants. It is straightforward to verify that $x^* = 0$. We restrict our attention to constant step sizes.
Define the composite iterate

$$w_k := \begin{pmatrix} x_k \\ x_{k-1} \end{pmatrix}.$$

Then the iteration can be written as

$$w_k = T w_{k-1}, \quad T = \begin{pmatrix} (1+\beta)I - \alpha Q & -\beta I \\ I & 0 \end{pmatrix}.$$

Iterating,

$$w_{k+1} = T^k w_1, \quad \Rightarrow \quad \begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} = T^k \begin{pmatrix} x_1 \\ x_0 \end{pmatrix}.$$

Hence, it suffices to bound $\|T^k\|$.

We will be using the following result from matrix analysis, which is very useful.
<u>Result.</u>

Let M be a square matrix and let $\rho(M)$ denote its spectral radius. Then for all $k \geq 0$, there exists a sequence $\{\varepsilon_k\}_{k\geq 0}$ with $\varepsilon_k \geq 0$ and $\varepsilon_k \to 0$ such that

$$\|M^k\| \leq (\rho(M) + \varepsilon_k)^k.$$

Thus, it suffices to bound $\rho(T)$.

Now we will try to calculate the eigenvalues of T. For that, we will show that T is similar to a block diagonal matrix. Hence, calculating the eigenvalues of the block diagonal matrix will be sufficient.
Let $Q = U\Lambda U^\top$ be the eigendecomposition of Q. Define the permutation matrix $\Pi \in \mathbb{R}^{2n\times 2n}$ by

$$\Pi(i,j) = \begin{cases} 1, & \text{if } i = 2j-1,\ 1 \leq j \leq n, \\ 1, & \text{if } i = 2(j-n),\ n+1 \leq j \leq 2n, \\ 0, & \text{otherwise.} \end{cases}$$

Then,

$$\Pi \begin{pmatrix} U & 0 \\ 0 & U \end{pmatrix}^\top T \begin{pmatrix} U & 0 \\ 0 & U \end{pmatrix} \Pi^\top = \begin{pmatrix} T_1 & 0 & \cdots & 0 \\ 0 & T_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_n \end{pmatrix},$$

where

$$T_i = \begin{pmatrix} 1+\beta-\alpha\lambda_i & -\beta \\ 1 & 0 \end{pmatrix}.$$

Thus, it suffices to compute the eigenvalues of each T_i.
For each $i = 1, \ldots, n$, the eigenvalues are roots of the quadratic equation

$$\nu^2 - (1+\beta-\alpha\lambda_i)\nu + \beta = 0.$$

Solving,

$$\nu = \frac{(1+\beta-\alpha\lambda_i)}{2} \pm \frac{\sqrt{(1+\beta-\alpha\lambda_i)^2 - 4\beta}}{2}.$$

If the discriminant is negative, the eigenvalues are complex and can be written as

$$\nu = a \pm ib,$$

where

$$a = \frac{1+\beta-\alpha\lambda_i}{2}, \quad b = \frac{\sqrt{4\beta - (1+\beta-\alpha\lambda_i)^2}}{2}.$$

The magnitude of the eigenvalues is

$$|\nu| = \sqrt{a^2+b^2} = \sqrt{\beta}.$$

Thus, for all $i = 1, \ldots, n$, the eigenvalues of T_i have magnitude is $\sqrt{\beta}$.

(b) From part (a), the eigenvalues satisfy

$$\nu^2 - (1+\beta-\alpha\lambda_i)\nu + \beta = 0.$$

For fast convergence, we want the eigenvalues to be complex with equal magnitude. This happens when the discriminant is negative:

$$(1+\beta-\alpha\lambda_i)^2 - 4\beta < 0.$$

This is equivalent to

$$(1-\sqrt{\beta})^2 < \alpha\lambda_i < (1+\sqrt{\beta})^2, \quad \forall i = 1, \ldots, n.$$

Since $\lambda_i \in [\ell, L]$, this implies

$$(1-\sqrt{\beta})^2 < \alpha\ell, \quad \alpha L < (1+\sqrt{\beta})^2.$$

Combining,

$$\frac{(1-\sqrt{\beta})^2}{\ell} < \frac{(1+\sqrt{\beta})^2}{L}.$$

Taking square roots,

$$\frac{1-\sqrt{\beta}}{1+\sqrt{\beta}} \leq \sqrt{\frac{\ell}{L}}.$$

Let $\kappa = \frac{L}{\ell}$. Then

$$\frac{1-\sqrt{\beta}}{1+\sqrt{\beta}} = \frac{1}{\sqrt{\kappa}}.$$

Solving for $\sqrt{\beta}$,

$$\sqrt{\beta} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}.$$

Thus,

$$\boxed{\beta = \left(\frac{\sqrt{L}-\sqrt{\ell}}{\sqrt{L}+\sqrt{\ell}}\right)^2}.$$

Now using $\alpha\ell = (1-\sqrt{\beta})^2$, we obtain

$$\alpha = \frac{(1-\sqrt{\beta})^2}{\ell}.$$

Substituting the value of β,

$$\boxed{\alpha = \frac{4}{(\sqrt{L}+\sqrt{\ell})^2}}.$$

(c) Using the result from matrix analysis, we have

$$\|T^k\| \leq (\rho(T)+\varepsilon_k)^k, \quad \varepsilon_k \to 0.$$

From part (b), the eigenvalues of T satisfy $|\nu| = \sqrt{\beta}$. Hence,

$$\rho(T) = \max_i |\nu_i| = \sqrt{\beta}.$$

Therefore,

$$\|T^k\| \leq (\sqrt{\beta}+\varepsilon_k)^k.$$

Using the relation

$$\begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} = T^k \begin{pmatrix} x_1 \\ x_0 \end{pmatrix},$$

we obtain

$$\left\| \begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} \right\| \leq (\sqrt{\beta}+\varepsilon_k)^k \left\| \begin{pmatrix} x_1 \\ x_0 \end{pmatrix} \right\|.$$

In particular,

$$\|x_k\| \leq (\sqrt{\beta}+\varepsilon_k)^k D_0, \quad \text{where } D_0 \text{ is a constant.}$$

Substituting

$$\sqrt{\beta} = \frac{\sqrt{L}-\sqrt{\ell}}{\sqrt{L}+\sqrt{\ell}} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$$

we obtain

$$\boxed{\|x_k\| \leq \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} + \varepsilon_k\right)^k D_0}.$$

Thus, the method converges linearly with rate

$$\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} = 1 - \frac{2}{\sqrt{\kappa}+1} \approx 1 - \frac{2}{\sqrt{\kappa}}.$$

Thus, we can write the convergence rate as

$$\rho \approx 1 - \frac{2}{\sqrt{\kappa}}.$$

<u>Note.</u>

The heavy ball analysis is elementary and powerful. The asymptotic rate is better than for Nesterov's method.

However, we need to note a few things:
(i) it depends on the knowledge of ℓ and L in order to choose α and β,
(ii) it does not extend neatly from quadratic functions to general nonlinear f, and
(iii) contraction cannot be established in the weakly convex case $\ell = 0$.

<u>Comparison (for strongly convex quadratic functions):</u>
Steepest descent: $\rho \approx 1 - \frac{2}{\kappa}$, Heavy-ball: $\rho \approx 1 - \frac{2}{\sqrt{\kappa}}$.